

B.E. II Year (Electronics & Communication Engineering), 2018-19

SEMESTER III EXAMINATION SCHEME

Branch Code	Subject Code	Subject	Lectures	Tutorials	Practicals	Contact Hours	Credits	Exam Hours	Marks		
									Theory	Pract & Sess	Total
A: Written Papers											
MA	201 A	Advanced Engineering Mathematics (EC/ECC/EEE) - I	2	-	-	2	2	3	100	-	100
EC	202 A	Network Theory (EC/ECC/EEE)	2	1	-	3	3	3	100	-	100
EC	207 A	Solid State Electronic Devices (EC)	2	1	-	3	3	3	100	-	100
EC	204 A	Electronic Circuit (EC) – I	3	1	-	4	4	3	100	-	100
EC	205 A	Digital Electronics (EC/ECC/EEE) – I	3	1	-	4	4	3	100	-	100
EC	206 A	Electronic Instruments & Measurements (EC)	2	1	-	3	3	3	100	-	100
Total (A)			14	5		19	19	-	600	-	600
B: Practicals and Sessionals											
EC	211 B	Network Laboratory (EC)	-	-	2	2	1	-	-	100	100
EC	212 B	Electronic Instruments & Measurement Lab (EC)	-	-	2	2	1	-	-	100	100
EC	213 B	Electronic Circuits Lab (EC) – I	-	-	2	2	1	-	-	100	100
EC	214 B	Digital Electronics Lab (EC) - I	-	-	2	2	1	-	-	100	100
EC	215 B	Logic System Design (EC) - I	-	-	2	2	1	-	-	100	100
EC	216 B	Computer Programming (EC)	-	-	2	2	1	-	-	100	100
Total (B)			-	-	12	12	6	-	-	600	600
Grand Total (A+B)			14	5	12	31	25	-	600	600	1200

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SEMESTER IV EXAMINATION SCHEME

Branch Code	Subject Code	Subject	Lectures	Tutorials	Practicals	Contact Hours	Credits	Exam Hours	Marks		
									Theory	Pract & Sess	Total
A: Written Papers											
MA	251 A	Advanced Engineering Mathematics (EC/ECC/EEE) - II	2	-	-	2	2	3	100	-	100
EC	252 A	R.F. Circuits & Systems	2	1	-	3	3	3	100	-	100
EC	254 A	Electronics Circuits (EC/EEE) – II	3	1	-	4	4	3	100	-	100
EC	255 A	Digital Electronics (EC/ECC/EEE) – II	2	1	-	3	3	3	100	-	100
EC	257 A	Industrial Electronics (EC/EEE)	3	1	-	4	4	3	100	-	100
EC	291A	OPEN ELECTIVE - I	3	-	-	3	3	3	100	-	100
Total (A)			15	4		19	19	-	600	-	600
B: Practicals and Sessionals											
EC	261 B	R.F. Circuits Lab (EC)	-	-	2	2	1	-	-	100	100
EC	263 B	Electronics Circuits Lab (EC) – II	-	-	2	2	1	-	-	100	100
EC	264 B	Digital Electronics Lab (EC) – II	-	-	2	2	1	-	-	100	100
EC	265 B	Logic System Design (EC) - II	-	-	2	2	2	-	-	100	100
EC	267 B	Industrial Electronics Lab (EC)	-	-	2	2	1	-	-	100	100
Total (B)					12	12	6	-	-	500	500
Grand Total (A+B)			15	4	12	31	25	-	600	500	1100
EC	200 E	Co-curricular Activities	-	-	-	-	0	-	-	-	100

*Choice based electives offered by Electronics & Communication Engineering Department in IVth semester:

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SEMESTER V EXAMINATION SCHEME

Branch Code	Subject Code	Subject	Lectures	Tutorials	Practicals	Contact Hours	Credits	Exam Hours	Marks		
									Theory	Pract & Sess	Total
A: Written Papers											
EC	304 A	Analog Communication Engineering (EC/ECC)	2	1	-	3	3	3	100	-	100
EC	306 A	Theory & Applications of Integrated Circuits (EC)	3	1	-	4	4	3	100	-	100
EC	309 A	Computer Organisation (EC/ECC/EEE)	2	1	-	3	3	3	100	-	100
EC	310 A	Radiation & Wave Propagation (EC) – I	3	1	-	4	4	3	100	-	100
EC	311 A	Feedback & Control Systems (EC)	2	1	-	3	3	3	100	-	100
EC	341A	Open Elective – II*	3	-	-	3	3	3	100	-	100
Total (A)			15	5		20	20	-	600	-	600
B: Practicals and Sessionals											
EC	313 B	Analog Communication Lab (EC)	-	-	2	2	1	-	-	100	100
EC	314 B	Radiation & Wave Propagation Lab (EC) – I	-	-	2	2	1	-	-	100	100
EC	315 B	Integrated Circuits Lab (EC)	-	-	2	2	1	-	-	100	100
EC	316 B	Electronics Workshop (EC)	-	-	2	2	1	-	-	100	100
EC	317 B	Control System Lab (EC)	-	-	2	2	1	-	-	100	100
Total (B)			-	-	10	10	5	-	-	500	500
Grand Total (A+B)			15	5	10	30	25	-	600	500	1100

***Choice based electives offered by Electronics & Communication Engineering Department in Vth semester:**

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SEMESTER VI EXAMINATION SCHEME

Branch Code	Subject Code	Subject	Lectures	Tutorials	Practicals	Contact Hours	Credits	Exam Hours	Marks		
									Theory	Pract & Sess	Total
A: Written Papers											
EC	352 A	Electronics Measurements (EC)	2	1	-	3	3	3	100	-	100
EC	353 A	Digital Communication Systems (EC/ECC)	3	1	-	4	4	3	100	-	100
EC	355 A	Radiation & Wave Propagation (EC) – II	2	1	-	3	3	3	100	-	100
EC	358 A	Microprocessor & Microcomputers (EC/ECC/EEE)	3	1	-	4	4	3	100	-	100
EC	361 A	Audio & Video Electronics (EC)	2	1	-	3	3	3	100	-	100
EC	391A	OPEN ELECTIVE – III*	3	-	-	3	3	3	100	-	100
Total (A)			15	5		20	20	-	600	-	600
B: Practicals and Sessionals											
EC	367 B	Audio & Video Lab (EC)	-	-	2	2	1	-	-	100	100
EC	362 B	Electronics Measurement Lab (EC)	-	-	2	2	1	-	-	100	100
EC	363 B	Digital Communication Lab (EC)	-	-	2	2	1	-	-	100	100
EC	364 B	Microprocessor Lab (EC)	-	-	2	2	1			100	100
EC	365 B	Radiation & Wave Propagation Lab (EC) – II	-	-	2	2	1	-	-	100	100
Total (B)			-	-	10	10	5	-	-	500	500
Grand Total (A+B)			15	5	10	30	25	-	600	500	1100
EC	300 E	Co-curricular Activities	-	-	-	-	0	-	-	-	100

***Choice based electives offered by by Electronics & Communication Engineering Department in VIth semester:**

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SEMESTER VII EXAMINATION SCHEME

Branch Code	Subject Code	Subject	Lectures	Tutorials	Practicals	Contact Hours	Credits	Exam Hours	Marks		
									Theory	Pract & Sess	Total
A: Written Papers											
EC	402 A	Telematics (EC)	3	1	-	4	4	3	100	-	100
EC	403 A	Advanced Communication Systems (EC/ECC/EEE)	3	1	-	4	4	3	100	-	100
EC	404 A	Advanced Computer Technology (EC/EEE)	3	1	-	4	4	3	100	-	100
EC	407 A	Radar & Navigational Aids (EC)	3	1	-	4	4	3	100	-	100
EC	408 A	Engineering Management & Economics (EC/ECC/EEE)	2	-	-	2	2	3	100	-	100
EC		Core Elective – I	2	1	-	3	3	3	100		100
Total (A)			16	5	-	21	21	-	600	-	600
B: Practicals and Sessionals											
EC	412 B	Telematics Lab (EC)	-	-	2	2	1	-	-	100	100
EC	413 B	Advanced Communication Systems Lab (EC)	-	-	2	2	1	-	-	100	100
EC	417 B	Advanced Computer Lab (EC)	-	-	2	2	1	-	-	100	100
EC	415 B	Project (EC) (Phase – I)*	-	-	2	2	-	-	-	-	-
EC	416 B	Seminar	-	-	1	1	1	-	-	100	100
Total (B)			-	-	9	9	4	-	-	400	400
Grand Total (A+B)			16	5	9	30	25	-	600	400	1000

* EC 415 B: A combined exam of Phase I & II shall be conducted at the end of VIII Semester.

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SEMESTER VIII EXAMINATION SCHEME

Branch Code	Subject Code	Subject	Lectures	Tutorials	Practicals	Contact Hours	Credits	Exam Hours	Marks		
									Theory	Pract & Sess	Total
A: Written Papers											
EC	451 A	Digital Signal Processing (EC/ EEE)	2	1	-	3	3	3	100	-	100
EC	452 A	Electronic Instrumentation (EC)	2	1	-	3	3	3	100	-	100
EC	457 A	Mobile Communication (EC)	2	1	-	3	3	3	100	-	100
EC	458 A	Computer Communication & Data Networks (EC/ECC/EEE)	3	1	-	4	4	3	100	-	100
EC	459 A	Microwave Engineering (EC/EEE)	3	1	-	4	4	3	100	-	100
EC		Core Elective – II	2	1	-	3	3	-	100		100
Total (A)			14	6		20	20	-	600	-	600
B: Practicals and Sessionals											
EC	461 B	DSP Lab (EC)	-	-	2	2	1	-	-	100	100
EC	462 B	Instrumentation Lab (EC)	-	-	2	2	1	-	-	100	100
EC	463 B	Microwave Lab (EC)	-	-	2	2	1	-	-	100	100
EC	465 B	Computer Communication & Data Networks Lab (EC)	-	-	2	2	1	-	-	100	100
EC	415 B	Project (EC) (Phase – II)*	-	-	2	2	1	-	-	200	200
Total (B)			-	-	10	10	5	-	-	600	600
Total (A+B)			14	6	10	30	25	-	600	600	1200
C: Others											
EC	476 C	Practical Training II (EC)	-	-	-	-	13	-	-	100	100
		Total (C)	-	-	-	-	13	-	-	100	100
Grand Total (A+B+C)			14	6	10	30	38	-	600	700	1300
EC	400 E	Co-curricular Activities	-	-	-	-	0	-	-	-	100

*Project (Phase II) is continuation of Project (Phase I) & combined examination shall be conducted.

List of Core Electives	
EC 481 A	Medical Electronics (EC)
EC 482 A	Design Principles of Communication Systems (EC)
EC 484 A	Solid State and IC Technology (EC)
EC 485 A	Computer and Microprocessor Control of Industrial Processes (EC)
EC 487 A	Entrepreneurship Development (EC)
EC 488 A	Reliability and Maintainability of Electronic Systems (EC)
EC 490 A	Robotics Engineering (EC)

List of Open Electives	
Name of subject	Semester
BCT 291 A Open Elective-I: Sustainable Architecture CE 291A Open Elective-I :Energy Efficient Building Design ChE 291 A Open Elective-I : Renewable Energy Sources CSE 291A Open Elective-I :Object Oriented Programming Through C++ CSE 292A Open Elective-I :Object Oriented Programming Through JAVA EE 291 A Open Elective-I : Industrial Applications of Electrical Drives Ma 291 A Open Elective-I :Mathematical Statistics For Engineers ME 291 A Open Elective-I:Renewable Energy Sources ME 292A Open Elective-I: Automobile Engineering MI 291A Open Elective-I: Tunneling For Engineering Projects PI 291A Open Elective-I: Manufacturing Science SE 291A Open Elective-I: Computer Oriented Numerical Analysis	Fourth Semester
BCT 341 A Open Elective-II: Traditional Indian Architecture CE 341A Open Elective-II: Non Urban Public Hygiene & Drinking Water ChE 341 A Open Elective-II : Petroleum Refining Technology CSE 341A Open Elective-II :Data Structures and Algorithms EE 341 A Open Elective-II : Optimization Techniques EE 342 A Open Elective-II : Artificial Intelligence Ma 341 A Open Elective-II :Mathematical Theory of Operations Research ME 342A Open Elective-II: Finite Elements Method MI 341A Open Elective-II: Application of GIS & Remote Sensing in Engineering SE 341A Open Elective-II :Structural Dynamics	Fifth Semester
BCT 391 A Open Elective-III: Climate Responsive Architecture CE 391A Open Elective-III: Ecosystem & Biodiversity ChE 391 A Open Elective-III : Nanotechnology CSE 391A Open Elective-III : WEB TECHNOLOGIES CSE 392A Open Elective - III : DATABASE MANAGEMENT SYSTEMS EE 391 A Open Elective-III: Soft Computing Techniques EE 392 A Open Elective-III: Energy Conservation Ma 391 A Open Elective-III: Advanced Numerical Analysis ME 391A Open Elective-III: Design Planning And Control Of Production System ME 392A Open Elective-III: Systems Design And Analysis MI 391A Open Elective-III: Project Environment Clearance PI 391A Open Elective-III: Quality Management SE 391A Open Elective-III :Finite Element Method	Sixth Semester
BCT: Building Construction Technology, CE: Civil Engineering, ChE: Chemical Engineering, CSE: Computer Science and Engineering, EC: Electronics & Communication, EE: Electrical Engineering, ME: Mechanical Engineering, MI: Mining Engineering, PI: Production & Industrial Engineering, SE: Structural Engineering, Ma: Mathematics	

List Of Open Electives Offered By The Department of Electronics & Communication Engineering Not applicable for ECE/ECC/EEE Students	
Name of subject	Semester
EC 291 A : Logic System Design	Fourth Semester
EC 341 A : Microprocessors & Microcontrollers	Fifth Semester
EC 391 A Open Elective-III: Electronic Instrumentation	Sixth Semester

SYLLABUS OF OPEN ELECTIVES-I

BCT 291 A Open Elective-I: Sustainable Architecture

3L

3 Hrs, MM 100

Concepts of sustainability : Energy and Global environment, Energy use and Climate change – Its impact, Types of Energy systems, Concept of Sustainability - Principles of conservation -synergy with nature, Bioregionalism - community basis shelter technology within bioregional patterns and scales, Ethical- environmental degradation. Sustainable planning & Design: Sustainable Development -Sustainable approach to site planning and design - site inventories- relationships between site factors - development impacts from one area of the site on the other areas - phasing of development - limits of change - Design facility within social and environmental thresholds Sustainable Building Materials and Construction : Properties, Uses and Examples of -Primary, secondary and Tertiary Sustainable Materials, Principles to improve the energy efficiency - siting and vernacular design, shade, ventilation, earth shelter, thermal inertia and air lock entrances. Techniques of sustainable construction - technologies, methods of effectiveness, and design synthesis – alternative materials and construction methods: solar water heating panels; photovoltaic electricity generation; use of local materials and on site growth of food, fuel and building materials.

Recycling and Reuse : Pre building, Building, Post building stages - Architectural Reuse, Waste prevention, Construction and Demolition recycling- Conservation of natural and building resources- Energy and material savings – types of wastes - Elimination of waste and minimize pollution- various Decomposing methods – Innovative reuse of various wastes Case Studies and Rating systems : Sustainable Development Case Studies: illustrated examples of the planning, development, and construction. Green architecture and various international rating systems for sustainability- EAM (UK), CASBEE (Japan), LEED (US), Green Star (Australia), etc. – Indian systems – TERI GRIHA rating, LEED India rating, IGBC

CE 291A Open Elective-I :Energy Efficient Building Design

3L

3 Hrs, MM 100

Environment and man, external environment and built environment, Built-environment – integrated approach. Climate: elements of climate, classification of climate, Micro-climate, site climate. Comfort: desirable conditions, thermal comfort factors, comfort indices, effective and corrected effective temp. Tropical summer index.

Thermal Design: heat loss from a building under steady state condition, heat gains due to solar radiation, steady state and cyclic conditions, Means of thermal control – mechanical, structural control, air infiltration into buildings by natural means, shape of buildings,-thermal cube, fabric heat loss, ventilation loss and volume.

Light & Lighting: illumination requirement, day-lighting, artificial lighting, energy conservation.

Noise Control: Sources of noise, means of control, control requirements, behaviour of sound in rooms, vibration & vibration control.

Building Services: Mechanical & electrical services in building, lifts, escalators.

ChE 291 A Open Elective-I : Renewable Energy Sources

3L

3 Hrs, MM 100

Sources of energy: Energy sources and their availability, renewable energy sources.

Energy from Biomass: Introduction, Biomass as a source of energy, Biomass conversion technologies, Biogas generation, classification of biogas plants, Biomass gasification.

Solar Energy: Sun and solar energy, solar radiation and its measurement, solar energy collectors, solar energy storage, Photovoltaic systems, Application of solar energy.

Wind Energy: Wind as an Energy source, Basic principles of wind energy conversion, Types of Wind machines, Components of wind energy conversion system, Performance of wind machines, application of wind energy.

Geothermal Energy: Introduction, Origin and distribution of geothermal energy, types of geothermal resources, Hybrid geothermal power plant, Application of geothermal energy.

Hydrogen energy: Introduction, Hydrogen production, Hydrogen storage, Hydrogen transportation.

Energy from the Oceans: Introduction, Ocean Thermal Electric Conversion (OTEC), Energy from Tides, Ocean Waves.

CSE 291A - Object Oriented Programming Through C++ (Open)

3L

3 Hrs, MM 100

A review of C. Concepts of object oriented programming using C++. Data types: elementary and derived data types, literals.

Operators and expressions: operators, association and precedence rules of operators, expressions using unary, binary and ternary operators.

Statements: declarations as statements, selection statements, iteration statements, goto statement, break statement, continue statement, return statement, try-catch block.

Functions: void functions, functions with return value, call by value and call by reference parameter passing, default parameters, recursive functions, inline functions.

Classes: classes, objects, friend functions, classes within a class, local classes, global classes, constructors, destructors.

Derived classes: single and multiple derivation of classes, multilevel and hybrid derivation of classes, constructors, destructors.

Polymorphism: function and operator overloading, virtual functions.

Streams: input and output of built-in data types, manipulators.

File streams: opening a file, accessing a file, closing a file.

Exceptions: catching exceptions, rethrowing the exception, standard exceptions.

Templates: defining a template, template instantiation, function templates, class templates.

Elementary case study of a object oriented database in C++.

CSE 292A - Object Oriented Programming Through JAVA (Open)

3L

3 Hrs, MM 100

Overview of object oriented concepts in JAVA.

Introduction – Java & internet, java applets and its applications, Java features like – security, portability, byte code, java virtual Machine, object oriented, robust, multi threading, architectural neutral, distributed & dynamic.

Data types and control structures, operators, arrays, Java methods and classes.

Inheritance of procedures and data, Packages and interface, exception handling, multi-threaded programming – thread priorities, synchronization, messaging, creating and controlling of threads. IO and applets.

String handling and various string functions.

Java utilities like java.lang, java.util and their uses, java.io, basics of networking using Java.

Java applets and their use, event handling, AWT and working with windows.

Introductory study of Java Beans, Servlets and JDBC.

EE 291 A Open Elective I: Industrial Application Of Electrical Drives

3L

3 Hrs, MM 100

Operating-Characteristics : Individual, group and collective drives, steady state individual and joint characteristics of electric motors and driven industrial units under different conditions of operation.

Transient Characteristics : Causes of transient conditions starting, braking, reversing, speed transition and sudden system changes. Forces and torques on the drives referred to a common reference shaft. General equation of motion, Accelerating and deaccelerating times. Starting and braking time and means of reducing.

Drives Control : Parameters characterizing speed control methods of electric drives, speed control of Industrial d.c. and a.c. motors under constant and varying torque and h.p. conditions.

Families of speed torque characteristics : Idea of manual and automatic control gears, Master-controller.

Motor Ratings : Continuous-short time and intermittent ratings, overload capacity. Effect of altitude, Motor heating and cooling curves. Equivalent current, power and torque. Selection motor for various duty cycles.

Permissible frequency of starting, features of load diagram construction. Load equalisation and use of fly wheels. Types of motor enclosures.

Illumination : Units of light, Point , linear and surfact sources. Laws of illumination. Candle power distribution, MSCP and reduction factor, Indoor lighting system and their classification. Contrast, glare, shadow and colour. Mounting height and spacing. General and local lighting Total lumen and point by point methods of calculations. Outdoor lighting distributor and protector fittings. Isolux diagram. Flood, gas, discharge and arc-lamp-working, characteristics and applications.

Electric Heating and Welding : Principles of electric heating. Direct and indirect resistance heating, lead baths and salt baths. Resistance oven convection and radiation ovens. Arc resistance and induction furnace, elements of operation, performance and power supply arrangements. temperature regulation of ovens and furnaces. Induction, high frequency and dielectric heating and their uses. Elementary study of different kinds of electric welding operation, Power supply for welding. Elements of Electrics Traction : Electric traction versus others System of electric traction for tramways, trolley buses, motor coach trains and locomotive hauled trains. Idea about suitability of electric motor for traction. Conductor rail and pantograph. meaning for multiple-unit operation.

Economics : Methods for economic selection of Industrial drives, loss factor and cost of losses, Effect of load factor. Power factor and factory diversity factor. Methods of power factor improvement and its economic limit. Economic calculations for illumination schemes Economic value of good lighting.

EC 291 A: LOGIC SYSTEM DESIGN

3L

3 Hours, 100 Marks

Number Systems & Codes: Binary, Octal and Hexadecimal number systems. Different numerical and alpha-numeric codes.

Basic Logic Circuits and Combinational logic: Positive and negative logic of OR, AND, NOT, NOR, NAND, Exclusive OR and Exclusive NOR gates. Boolean algebra, Boolean functions and expressions. Simplification and minimization techniques, K-map and tabular methods. Design of minimal combinational systems and realization.

Arithmetic circuits: Design and realization of Digital comparator, half and full adders, parallel and serial binary adders, half and full subtractors.

Miscellaneous sub systems: Encoders, decoders and code converters. Parity generator and parity checking circuits. Multiplexers and demultiplexers.

Sequential logic: Storage devices and sequential sub systems. Introduction to synchronous and asynchronous sequential systems. Mealy and Moore circuits.

Synchronous sequential systems: Introductory examples, memory elements and their excitation functions. Synthesis of synchronous sequential circuits. Analysis and design of synchronous sequential circuits.

Asynchronous sequential circuits: Fundamental mode circuits. Analysis procedure. Circuits with latches. Design of pulse mode asynchronous sequential circuits. Problems in asynchronous circuits - races and hazards.

Ma-291 A Open Elective-I :Mathematical Statistics for Engineers

3L

3 Hrs, MM 100

Theory of probability : Theoretical probability distribution (Binomial, Poisson and Normal).

Correlation and Regression Analysis : Karl-Pearson's coefficient , Spearman's coefficient, Regression analysis of two variables system.

Sampling Theory : Test of significance, Large sample tests for mean and proportions. χ^2 (chi-square) , t and F Test of significance for Small sample.

Theory of attributes: association and independence of attributes, coefficient of association.

Index Number: Various types of index numbers, construction of index number of prices, fixed base and chain base methods

ME- 291A Open Elective-I: Renewable Energy Sources

3L

3 Hrs, MM 100

Principal types of fossil fuel fired power plants and their effects on livestock and environment; Concepts of NCES, Criteria for assessing the potential of NCES, Limitations of RES.

Solar Energy - Solar radiation data, solar energy conversion into heat, Analysis of Flat plate and Concentrating collectors, Testing procedures, Paraboloid Dish, Central Receiver; concept of collector tracking, energy storage systems; Solar thermal systems for residential water heating, Solar Pond, industrial process heating and power generation. Photo voltaic: p-n junctions, I-V characteristics of solar cells, Calculation of energy for photovoltaic power generation; Battery Characteristics, DC Power Conditioning Converters, AC Power Conditioning - Inverters.

Wind Energy: Energy available from wind, General formula, Lift and drag. Basis of Wind energy conversion, Effect of density, Frequency variances, Angle of attack, Wind speed, Determination of torque coefficient, Principle of Operation of wind turbines, types of wind turbines and characteristics, Generators for Wind Turbines, Control strategies.

Biomass and Biofuels: Conversion routes- combustion, pyrolysis of biomass to produce solid, liquid and gaseous fuels; Constructional details of gasifier; Aerobic and anaerobic digestion, Biofuels and their production; biofuels, Biomass systems for thermal applications and power generation.

Geothermal Energy: Definition and classification of resources, typical geothermal gradient; Dry, flash and binary steam systems; Utilization for electricity generation and direct heating, Wellhead power generating units. Basic features: Atmospheric exhaust and condensing, Exhaust types of conventional steam turbines.

An overview of other renewable devices- Fuel cells: principle, types, applications; Ocean thermal energy conversion (OTEC), Thermoelectric, MHD, Wave energy, Tidal energy, etc.

Economic Viability: Calculation of the cost of energy supply from renewables, Payback period, Carbon footprints; Comparison with conventional fossil fuel driven systems in terms of costs and emissions; Calculation of carbon dioxide reduction and incremental costs for renewable options.

ME 292 A Open Elective-I: Automobile Engineering

3L

3Hrs, MM 100

Power Unit: Automobile engine types, classification; Engine parts: cylinder head, block and crank case, piston and rings; Carburation, fuel injection, valve operation; Fuel combustion, mechanical power and engine performance characteristics; Engine cooling and thermal stresses in parts,.

Chassis and Suspension: Load on frame, general considerations for strength and stiffness, engine mounting; Dampers, leaf and coil springs, various arrangements of suspension systems.

Transmission System: Clutches, flywheels, torque convertors; Gear-box: simple, synchromesh and overdrive; Type of universal joints, propeller shaft, differential; Rolling, air, gradient resistances and propulsive power calculation.

Steering: Steering geometry, Ackermann and Davis steering mechanisms; Telescopic steering; Steering shaft, gear-box, linkages, steering angles, front and rear axles; Vehicle longitudinal, static and dynamic balancing and electronic stability; Power steering: types and mechanism; Effect of caster, camber, toe-in and toe-out on tyre wear.

Brakes and Tyres: Servo-action, brake components; Bendix and Gerling system lock-head, hydraulic, vacuum, air and power brakes, and retarders; Pneumatic and tubeless tyres;

Features of a Modern passenger Car: Introduction to ABS, Front and side air bags, EBD, Climatizer, ESP, night-vision dashboard system; sun-roofing, collision warning system, Hybrid cars.

MI 291A Open Elective-I: Tunneling for Engineering Projects (MI)

3L

3 Hrs, MM 100

Tunneling: Introduction about tunnels, functions, advantages and disadvantages of tunnels compared to open cuts, Criteria for selection of size and shape of tunnels, consideration in tunneling, geological investigation, tunnel alignment, tunnel shafts, pilot tunnels. Advantages of twin tunnels and pilot tunnels, portals and adits.

Conventional Method of Tunneling: Drilling, Blasting, Loading and Transport of Muck, Supports, Ventilation, Drainage, and Equipments. Drivage work in varying ground conditions using conventional methods

Fast Tunneling: Dill jumbos, trackless mucking and transportation units. Tunnel boring machine

Tunneling in Soft Ground: General characteristics of soft ground, shield methods, needle beam method and NATM method of tunneling in practice.

Tunneling (rock bolting and guniting), Safety measures, Ventilation in tunneling, Lighting, Drainage.

PI 291A Open Elective-I: Manufacturing Science

3L

3 Hrs, MM 100

History and introduction to science of basic manufacturing processes and its classification.

Primary manufacturing processes: Introduction to liquid state forming process (casting), solid state forming process (drawing, extrusion, rolling, forging and other sheet metal working) and powder state forming process (powder metallurgy).

Secondary manufacturing processes: Introduction to material removal processes

Conventional Machining processes (basic machining operations performed of lathe, shaper, milling, drilling and grinding machine).

Introduction to basic metal joining processes (welding, brazing, soldering and mechanical fastening).

Non conventional machining processes (Basic introduction, classification, need for their development, characteristics and their industrial applications).

SE 291 A open elective-I: Computer oriented Numerical Analysis

3L

3 Hrs. MM 100

Error Analysis:- Approximations and errors, Round off errors. Roots of Equations:- Bisection method, Newton – Raphson method. Curve Fitting:- Linear Regression, Least Square Ft, Co-relation. Interpolation:- Linear & Quadratic, Newton's & Lagrange's polynomials. Numerical Differentiation:- Forward / Backward / Centered F.D. method. Numerical Integration:- Trapezoidal rule, Simpson's rule. Solution of simultaneous Linear algebraic equations.

SYLLABUS OF OPEN ELECTIVES-II

BCT 341 A Open Elective-II: Traditional Indian Architecture

3L

3 Hrs, MM 100

To provide theoretical knowledge base on the uniqueness of Indian traditional Architecture principles, the meaning of space, the manifestation of energy, the selection of site and how integration of built form with site happens at metaphysical level based on articulation of celestial grid.

To introduce the principles of Vastu and relationship between building and site. To familiarize the students with the units of measurement in traditional architecture. To introduce concepts of orientation and cosmogram according to the Vasthu Purusha Mandala. To study the detailing and design of various building components and their material and method of construction.

Vastu - its definition and classification - Relationship to earth.. Features of good building site - good building shapes - macro, micro, enclosed and material spaces - relationship between built space, living organism and universe - impact of built space on human psyche.

Orientation of building, site, layout and settlement - positive and negative energies - importance of cardinal and ordinal directions - The celestial grid or mandala and its types. The Vastu Purusha Mandala and its significance in creation of patterns, and lay-outs, Types of lay-outs. Simple design of residential buildings.

Building heights -Base and basement - wall and roof specifications - column and beam designs - Pitched roof and domical roofs - significance of pyramid.

Use of wood, stone, metal, brick and lime - marking technology, corbelling technology, jointing technology - foundations for heavy and light structures - Landscaping in and around buildings - Aesthetics in Indian Architecture.

CE 341A Open Elective-II: Non Urban Public Hygiene & Drinking Water

3L

3 Hrs, MM 100

Communicable disease: Disease and immunity, communicable disease sources, mode of transfer. Control of communicable disease.

Fly and mosquito control: Life cycle of flies and mosquitoes. Various methods of fly and mosquitoes control.

Milk and food sanitation: Essential of dairy farm and cattle shed sanitation. Tests for milk and dairy products.

Food epidemic, food poisoning. Botulism. Rural sanitation, village latrines, aqua privies, storm water and sullage problems, animal waste, methods of composting. Biogas collection and disposal of refuse, solid waste management through vermicomposting.

Septic tank (only salient features), percolation pits, sub surface disposal.

Rural water supply: Importance of village community in India, conditions of Indian villages with special regards to economic, social and health aspects. Quality of water needed for village community, sources of water for village water supplied, domestic roof water harvesting. Types of wells of sanitary aspects in well construction.

Disinfections of wells. Different types of pumps used for village wells. Operation and maintenance of pumps,

water borne diseases. Quality of water, human and cattle population and their water requirement. Rate of water supply. Standards of potable water. Rain water storage.

Treatment of water: Disinfection, desalination, Defluoridation, distribution of water.

ChE 341 A Open Elective-II: Petroleum Refining Technology

3L

3 Hrs, MM 100

Origin occurrence of petroleum, Formation and Evaluation of Crude Oil. Testing of Petroleum Products. Petroleum refining processes, general processing, topping and vacuum distillations. Thermal cracking in vapor, liquid and mixed phase. Overview of Refinery Products

Catalytic cracking - Houdry fixed bed, fluidized bed, T.C.C. Houdry flow etc. Catalytic reforming - conversion of petroleum gases into motor fuel with special reference to alkylation, polymerization, hydrogenation and dehydrogenation.

Treatment Techniques: Removal of Sulphur Compounds in all Petroleum Fractions to improve performance, Destruction of Sulphur Compounds and Catalytic Desulphurization, Solvent Treatment Processes, Dewaxing, Clay Treatment and Hydrofining.

Production of aviation gasoline, motor fuel, kerosene, diesel oil, tractor fuel and jet fuel, hydrodesulfurisation, Lubricating oil manufacture, Petroleum waxes and asphalts.

Octane number, Cetane number, Diesel index, their determination and importance Storage of petroleum products: tanks, bullets, special types of spheres etc. Transportation of petroleum products: road, rail, sea and pipeline; Importance of pipeline transportation.

CSE 341A Open Elective-II data structures and algorithms (Open)

3L

3 Hrs, MM 100

Introduction to data structure, String storage representation and manipulation. Markov algorithm and primitive data structures.

Concepts of non primitive data structures. Linear data structure. Array, stack, queue, their applications and implementations using sequential storage representation and linked representation.

Linear linked list, double linked list, circular linear linked list and generalized lists and applications.

Concept of non-linear data structures, Tree, graph, set and their representation, Binary Tree, Threaded tree, different techniques of tree traversal, breadth first search, depth first search, application of tree and graph such that Polish notation, concepts of heap.

Sorting, searching algorithms and comparative study of different sorting and searching techniques such that selection sort, heap sort, bubble sort, quick sort, merge sort and radix sort. Linear search and binary search, hashing. External sorting.

Time and space complexity of the algorithms – Big-O, θ , Ω , and small-o, Asymptotic complexity, Upper and Lower bound time and space trade offs.

EE 341 A Open Elective-II: Optimization Techniques

3L

3 Hrs, MM 100

Introduction to Optimization : Historical Development, Engineering application of optimization, Statement & Classification of optimization problems, Classical optimization techniques for single & multiple variable functions.

Linear Programming : Introduction, application, standard form, Basic Solutions, Simplex method, Revised Simplex method, Duality, Transportation problem, Carmarkar's method.

Nonlinear programming : Unconstrained Optimization, Introduction, Fibonacci method, Golden section search, Gradient method, Newton's method, Quasi Newton method.

Dynamic Programming : Introduction, Multistage decision process, Concept of optimization & principle of optimality, Computational procedure in dynamic programming.

Advanced topics in optimization : Introduction, Separable programming, Multi objective optimization, Calculus of variation.

Books :

Engineering Optimization – S. S. Rao, New Age International Publishers.

An introduction to optimization – Er. K. P. Chong, S. H. Zak (Wiley Slident Edition)

Operations Research : An international – H. A. Taha (PHI)

Introduction to operation research – Hiller F.K. & Lieberman (TMH)

EE 342 A Open Elective-II: artificial intelligence (e)

3L

3 Hrs, MM 100

Introduction: Economics for Electrical Engineering, concept of physical efficiency and financial efficiency of electrical goods and services supply and demand, Elasticity. Necessities and luxuries, free competition, monopoly, law of diminishing returns.

Interest and Depreciation: Interest rates and equivalence, annuities and various factors, concept of depreciation in utilizing electrical energy, economic life of electrical machines, salvage value, various methods of depreciation calculations, equivalent capital recovery depreciation.

Economical choice of Electrical Apparatus: Motors, transformers, Economical choice between synchronous motors and Induction motor running them simultaneously.

Comparison of Alternatives: Basic economic study patterns, annual cost, capitalized cost, present worth, rate of return, Increment investment, pay back and benefit to cost ratio methods and their respective fields of applications.

EC 341 A: MICROPROCESSORS AND MICROCONTROLLERS

3L

3 Hrs., MM : 100

Microprocessor Architecture: Architecture of 8-bit 8085 Microprocessor; instruction set and addressing modes. Assembly language programming of Intel's 8085 Microprocessor. Introduction to assemblers.

Microprocessor interfacing: Interfacing of address, data and control buses, Memory and I/O devices, Interrupt and DMA for 8085 microprocessor.

Introduction to Microcontrollers: Architecture and instruction set of MCS-51 series of microcontrollers. Applications of Microcontrollers.

16 and 32-bit Microprocessors: CPU architecture, addressing modes and features of 16 and 32 bit microprocessors – 8086. Salient features of 80286, 80386, 80486 and Pentium series microprocessors.

Bus standards: Introduction to Multibus , VME, RS-232-C, IEEE 488, PCI, USB, RS 422 and 485.

Ma 341 A Open Elective-II :Mathematical Theory of Operations Research

3L

3 Hrs, MM 100

Linear programming problems-Simplex Method, two phase method, Duality of LPP.

Theory of games: Competitive strategies, minimax and maximin criteria, two person zero-sum games with and without saddle point, dominance .

Inventories: Single item deterministic inventory models with finite and infinite rates of replenishment, economic lot-size model with known demand

Replacement problems: Replacement of item that deteriorate, replacement of items that fail completely, group replacement policy, individual replacement policy

Queuing theory-Ques with Poisson input and exponential service time, the queue length, waiting time and busy period in steady state case, model with service in phase, multiserver queueing models.

ME 342A Open Elective-II: Finite Elements Method

3L

3 Hrs, MM 100

Introduction: Basic concept of Finite element method; Rayleigh-Ritz and weighted residual method of variational approximation, Numerical Solution of equilibrium problem by Gaussian elimination.

Finite Element Analysis of One-dimensional Problem; Basic Concepts, derivation of elements equations, connectivity of elements, imposition of boundary conditions, Solution of equations, Application in One dimensional problem of Solid mechanics and heat transfer.

Finite Element Analysis of Two Dimensional Problem: Single variable problems: finite element discretization, interpolation, function, numerical integration and modeling considerations for triangular, rectangular, Quadrilateral, Isoparametric and Plane frame elements, Evaluation of equation and their solutions, Application in Two Dimensional Problem of Solid mechanics, Heat Transfer and Eigen value problems.

MI 341A Open Elective-II: Application of GIS & Remote Sensing in Engineering

3L

3 Hrs, MM 100

Remote Sensing: Introduction to Remote Sensing, Terminology in Remote Sensing, Types of Remote sensing, advantage and disadvantage of remote sensing data, Electromagnetic radiation atmospheric. Windows remote sensing platforms and sensors systems, path-row referencing system, remote sensing data product, procedure for obtaining satellite data. Hardware and software related to remote sensing.

Different types of platforms, sensors and their characteristics, Orbital parameters of a satellite, Multi concept in remote sensing.

Image Interpretation and analysis: Elements of visual image interpretation, Digital image pre-processing, radiometric correction, geometric correction, resolution of remote sensing data, image enhancement, contrast enhancement, spatial filtering band rationing image classification supervised and unsupervised classification, remote sensing applications in forestry, geology, hydrogeology, Land use and land cover mapping.

Principles of interpretation of aerial and satellite images, equipments and aids required for interpretation, ground truth – collection and verification, advantage of multiband and multiband images. Digital image Processing concept.

Geographic Information System (GIS): Fundamental of GIS: Basis concept including definition and history of GIS, Essential Elements of GIS, Uses and users of GIS, General GIS Applications, Geodesy, Grids, Datum's and projection systems, GIS Data structure, Data Formats, GIS layers and Digitization overview of GPS and its application, Hardware and software related to GIS.

Raster and vector Based GIS: Raster based GIS, Definition of Raster Based GIS, Spatial Referencing Definition and Representation of Raster Data. Vector based GIS, Definition and concept of vector based GIS, Data structure, Data Capture and Basic operations of spatial analysis, advantages and disadvantage in raster and vector based GIS, Introduction to network in GIS, GIS Project Planning Management and Implementation.

Application of GIS :in Map revision, Land use, Agriculture, Forestry, Archaeology, Municipal, Geology, water resources, soil Erosion, Land suitability analysis, change detection, Use of GIS in Mining.

SE 341A Open Elective-II :Structural Dynamics

3L

3 Hrs, MM 100

Vibrations of single degree of freedom system, sources of vibration, Types of vibration, Degree of freedom, spring action and damping, equation of motion of single degree of freedom system, undamped system of single degree of freedom, combination of stiffnesses, damped system of single degree of freedom, dry friction, damping forced vibration of damped system, introduction to multi degree freedom system.

SYLLABUS OF OPEN ELECTIVES-III

BCT 391 A Open Elective-III: Climate Responsive Architecture

3L,

3 Hrs, MM 100

Understanding Climate and its impact on architectural design, fundamentals of climatology and environmental studies.

Introduction – Elements of Climate, measurement and representation of climatic data. Classification of climate, major climatic zones of India.

Thermal Comfort: Effect of climatic elements on Thermal comfort; indices for Thermal comfort Thermal performance of building elements: Thermal and physical properties of building materials and their effect on indoor environments.

Natural ventilation: Functions, effects of openings and external features on internal air circulation. Design considerations for achieving natural ventilation.

Sun path diagram, use of solar charts, types of shading devices

Day light factor: components, design considerations for indoor spaces

Micro Climate: factors and effects

Construction techniques for improving thermal performance of walls and roofs. Passive cooling techniques: traditional and contemporary

Design considerations for buildings and settlements in tropical climates with special reference to hot-dry, warm-humid and composite climates; Mahoney Tables.

Exercises:

Design of shading devices.

Layout of Residence for hot - dry, warm-humid and composite climates.

CE 391A Open Elective-III: Ecosystem & Biodiversity

3L,

3 Hrs, MM 100

Concept of an ecosystem, structure & function of ecosystem, Bio-Geo chemical cycles (Hydrological, carbon, oxygen, nitrogen, phosphorus & sulphur cycle), energy flow in ecosystem, food chain

Major ecosystems (Description only) : Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystem, Riverine and stream ecosystem, Marine ecosystem, Estuarine ecosystem.

Biodiversity : Definition and its importance. Biodiversity at global, national & local level. Hot spots of biodiversity, Threats to biodiversity & causes of biodiversity loss. Conservation of biodiversity.

Value of biodiversity: Consumptive use, productive use, social value, ethical value, aesthetic value& optional value.

Bio-geographical classification of India. India as mega- diversity nation

ChE 391 A Open Elective-III: Nanotechnology

3L,

3 Hrs, MM 100

Introduction to Nanotechnology: Introduction to nanotechnology and materials, Nanomaterials, Introduction to nano sizes and properties comparison with the bulk materials, different shapes and sizes and morphology.

Fabrication of Nanomaterials: Wet Chemical Synthesis Methods, Colloidal Nanoparticles Production, Sol Gel Methods, Microwave and Atomization, Gas phase Production Methods : Chemical Vapour Depositions.

Kinetics at Nanoscale: Nucleation and growth of particles, Issues of Aggregation of Particles, Layers of surface Charges, Zeta Potential and pH.

Carbon Nanomaterials: Synthesis of carbon buckyballs, List of stable carbon allotropes extended fullerenes, metallofullerenes solid C60, bucky onions nanotubes.

Nanomaterials characterization: Instrumentation Fractionation principles of Particle size measurements, Particle size and its distribution, XRD, Zeta potential Microscopies SEM, TEM, Atomic Forced Microscopy, Scanning and Tunneling Microscopy

Applications in Chemical Engineering: Self-assembly and molecular manufacturing : Surfactant based system Colloidal system applications, ZnO,TiO₂, Silver Nanoparticles Functional materials Applications, Production Techniques of Nanotubes, Carbon arc, bulk synthesis, commercial processes of synthesis of nanomaterials, Nanoclay, Commercial case study of nano synthesis - applications in chemical engineering.

CSE 391A Open Elective-II: Web Technologies (Open)

3L,

3 Hrs, MM 100

Understanding Microsoft .NET Framework and ASP.NET, Creating components in Visual C#. CLR, Framework Class Library, Undocumented Types.

Programming the .NET Framework, Common Types, Math, Strings, Collections, Regular Expressions. Core Types, Serialization, Remoting, Graphics, Rich Client Applications, Globalization, Configuration, Advanced Component Services.

Multithreading, Thread Synchronization, Inter-thread Communication and Monitor. Delegates & Events. Validating User Input.

Creating a Connection to the Database, Displaying a DataSet, List-Bound Control, Paging and Selection, DataGrid Control, Accessing Data with DataReaders and SqlDataReader. Overview of Stored Procedures.

Managing State, State management, Application and Session Variables, Cookies and Cookieless Sessions. Configuring, Optimizing. Using the Cache object.

Reading and Writing XML Data, Overview of XML Architecture, DataSet Object, XML Web Server Control, Reading, Transforming, and Displaying XML, Nested Data. Creating an XML Web Service.

Securing a Microsoft ASP.NET Web Application, Web Application Security Overview, Windows-Based Authentication, Forms-Based Authentication, Passport Authentication, Registering New Users, Permitting Users to Sign Out.

AJAX.NET Architecture, Working with AJAX Pro and Controls, Accordion, Calendar, CascadingDropDown, CollapsiblePanel, Filtered TextBox, Numeric Up Down, Modal Popup, Popup Control. Page_Load Event and Click Event Procedure, Adding server controls to an ASP.NET Web Form basics, handling text and numbers.

CSE 392A Open Elective-II: Database Management Systems (Open)

3L

3 Hrs, MM 100

Purpose of data base system, data abstraction, data models, data independence, data definition language, data manipulation language, data base manager, data base administrator, data base users, overall system structure.

E-R Models, entities and entity sets, relationships and relationship sets, attributes, mapping constraints, keys, E-R diagrams, reducing E-R diagrams to tables, generation, aggregation, design of an E-R data base scheme

Basic concept of object oriented model, New database applications, object structure, class hierarchy, multiple inheritance, object identity, object containment, physical organization, object oriented queries, scheme modification.

File and system structure, overall system structure, file organization, logical and physical file organization, sequential, random, hierarchical, inverted, multilist, Indexing and hashing, B-tree index files

Introduction to distributed database. Introduction to SQL Query and SQL joins.

EE 391 A Open Elective-III: Soft Computing Techniques

3L

3 Hrs, MM 100

Soft Computing : Introduction of soft computing, soft computing vs. hard computing, various types of soft computing techniques, applications of soft computing.

Artificial Intelligence : Introduction, Various types of production systems, characteristics of production systems, breadth first search, depth first search techniques, other Search Techniques like hill Climbing, Best first Search, A* algorithm, AO* Algorithms and various types of control strategies. Knowledge representation issues, Propositional and predicate logic, monotonic and non monotonic reasoning, forward Reasoning, backward reasoning, Weak & Strong Slot & filler structures, NLP.

Neural Network : Structure and Function of a single neuron: Biological neuron, artificial neuron, definition of ANN, Taxonomy of neural net, Difference between ANN and human brain, characteristics and applications of ANN, single layer network, Perceptron training algorithm.

Fuzzy rule base system : fuzzy propositions, formation, decomposition & aggregation of fuzzy rules, fuzzy reasoning, fuzzy inference systems, fuzzy decision making & Applications of fuzzy logic.

EE 392 A Open Elective-III: Energy Conservation

3L

3 Hrs, MM 100

Elements of Energy Conservation and Management : General energy problem, Sector wise Energy consumption, demand supply gap, Scope for energy conservation and its benefits, Energy conservation Principle Maximum energy efficiency, Maximum cost effectiveness. Mandatory provisions of EC act Features of EC act Standards and labeling, designated consumers, Energy Conservation Building Codes (ECBC). Energy management concept and objectives Initializing Planning, Leading, Controlling, Promoting, Monitoring and Reporting, energy management programmes. **Energy Conservation Approaches In Industries :** energy saving opportunities in electric motors Benefits of Power factor improvement and its techniques Shunt capacitor, Synchronous Condenser etc., Effects of harmonics on Motors, and remedies leading to energy conservation Energy conservation by VSD Methods and techniques of energy conservation in ventilation and air conditioners compressors pumps, fans and blowers Area Sealing, Insulating the Heating / cooling fluid pipes, automatic door closing Air curtain, Thermostat / Control Energy conservation in electric furnaces, ovens and boilers lighting techniques Natural, CFL, LED lighting sources and fittings **Energy Conservation in Power Generation, Transmission and Distribution :** Performance improvement of existing power plant: cogeneration, small hydro, DG Set. Demand side management Load response programmes Types of tariff and restructuring of electric tariff Technical measures to optimize T and D losses.

EC 391 A: ELECTRONIC INSTRUMENTATION

3L

3 Hours, 100 Marks

Transducers: Construction, characteristics and circuits for common types of resistive, capacitive, inductive, magneto-structive; piezo-electric. Photo-electric and thermo-electric transducers for measurement of process physical variables. Various sensing elements and transducers for measurement of Force, Pressure, Humidity, Moisture, strain, Velocity, Acceleration and pH. Inductive and Capacitive proximity switches. Physical and electrical loading of and by the transducer Systems.

Signal Conditioning: Analog and digital signal conditioning for instrumentation. Objectives of DAS, components of analog DAS and digital Data acquisition system, digital data recording system, multi channel DAS, modern digital acquisition system.

Electronic Displays: Principle of LED numeric, matrix and alpha-numeric displays, flat panel CRT, LCD, electro-luminiscent and electrophoretic and touch screen displays.

Ma 391 A Open Elective-III: Advanced Numerical Analysis

3L

3 Hrs, MM 100

Solution of Algebraic and Transcendental Equations: Newton-Raphson method for real multiple roots, for complex roots and for system of non-linear equations; Synthetic Division, Birge-Vieta Method.

Solution of simultaneous Linear Equations and Eigen Value Problems: Direct methods: Gauss-elimination, Gauss-Jordan, Iterative Methods: Jacobi iteration, Gauss-seidel iteration and Successive Relaxation method. Eigen value Problems: power method

Curve fitting and Function Approximation: Chebyshev approximations, Chebyshev Expansion, Chebyshev Polynomials. Economization of Power Series.

Numerical Solution of Partial Differential Equations: Finite difference Approximation to partial derivatives.

Solution of Laplace and poisson equations, Solution of one and two dimensional heat and wave equation by the method of separation of variables.

ME 391A Open Elective-III: Design Planning and Control of Production System

3L

3 Hrs, MM 100

Production Planning: Planning horizon, product exploring, Make and buy decisions, operations planning, demand forecasting, conversion of forecast into production goal.

Scheduling: Operation sequencing and balancing, Scheduling for mass production and job order production, MRP, ERP.

Inventory System: Cost factors relevant to operations and inventory control, EOQ with shortages and uniform production, quantity discount.

Project Planning and Control: Network control, control cost consideration and optimization, Resource allocation and levelling, Aggregate production planning, decision rules.

Supply Chain Management:: Strategic framework of Supply chain – meaning, scope and performance of supply chain, supply chain drivers and obstacles. Role of e-business in a supply chain.

ME 392A Open Elective-III: Systems Design and Analysis

3L

3 Hrs, MM 100

System Fundamental Concept: System definition, systems approach, Classification- General Systems, Discrete Systems, Controlled systems.

Procedure for Engineering a system: Defining system objective, formulation of objective criteria, Development of system alternatives.

Systems Optimization: Formulation of system, Design problem and application of search methods, Linear programming and dynamic programming for optimum solutions.

System Schedule: Time models, resource allocation, Time cost trade-off, system cost economic flow graph.

MI 391A Open Elective-III: Project Environment Clearance

3L

3 Hrs, MM 100

Brief introduction of Environment Protection Act 1986 and other relevant legal provisions applicable to get environment clearance in India.

Impact of major engineering projects on various components of the environment: Socio-Economy, Land, Water, Air, Noise and others.

Preparation of Environment management plan: Public hearing, collecting baseline data, Environment impact assessment and predication, Environment management plan, environment monitoring and management.

PI 391A Open Elective-III: Quality Management

3L

3 Hrs, MM 100

Introduction: History of Quality, Objectives , importance and need of quality, Contributions of Quality Gurus- Juran, Deming, Crosby, Feigenbaum, Ishikawa, Taguchi etc., Impact of Quality on business performance.

Process and Statistical Quality Control: Quality System; Quality control techniques; Process capability; Control Charts- Theory of control charts, control limits and specification limits, Control charts for variables-X R Charts, Control Charts for attributes p, np, c and u charts.

Acceptance Sampling : Fundamental concepts of acceptance sampling; OC Curves; Single , Double and multiple sampling;

Quality Management: Introduction to Quality management; quality control and quality assurance; Quality control tools; cost of quality and cost of poor quality.

ISO 9000: ISO 9000 series; terminologies; need for ISO 9000 certification; basic procedure and work instructions; steps in ISO 9000 registration; Internal and third party audit for registration; Clauses of ISO 9000-2000 .

SE 391A Open Elective-III :Finite Element Method

3L

3 Hrs, MM 100

Introduction to Finite Element Method, Basic Concept of Finite Element Method, Analysis of continuum:- Structural, thermal, Potential etc., Finite Element Analysis of an elastic continuum:- Displacement approach, Direct Formulation, Energy Integral, Co and C1 continuity, Convergence criteria.

Elements:- Types and Properties. Conforming and Non conforming.

Shape Functions:- Lagrangian and Serendity family for one and two dimensional cases.

Pascal triangle, Super / Sub and Iso parametric elements.

Steps in Finite Element Analysis of an elastic continuum.

SECOND B E. EXAMINATION

THIRD SEMESTER

Ma 201 A (EC/ECC/EEE) - Advanced Engineering Mathematics I

2L

3 Hours, 100 Marks

Section A

Differential equations : Simultaneous differential equations, Total differential equations, Partial differential equations of first order, Charpit's method, Linear homogeneous partial differential equations with constant coefficients, Second order partial differential equations : Monge's method for the equation of type

$$Rr + Ss + Tt = V$$

Solution of Wave, Heat and Laplace equations using separation of variables method.

Section B

Complex Analysis : Analytic function, Harmonic function, Construction of an Analytic function, Complex integration: Cauchy's integral theorem, Cauchy's integral formula, Derivative of Cauchy's integral formula.

Taylor's and Laurent's series expansion of complex functions.

Cauchy's residue theorem and its application to evaluate the contour integrals of type $\oint_0^{2\pi} f(x)dx$ and $\int_{-\infty}^{+\infty} f(x)dx$

Transformations: conformal and bilinear transformations.

Section C

Probability and Statistics: Theorems of probability and their application, Binomial, Poisson and Normal probability distribution. Correlation and Regression analysis of two parameters.

Note: Candidates are required to attempt FIVE questions in all, selecting at least one from each Section.

EC 202 A - NETWORK THEORY (EC/ECC/EEE)

2L, 1T

3 Hours, 100 Marks

Network Equations: Topology incidence, cut-set and tie-set matrices. Mesh and nodal analysis of networks with independent and dependent sources. Duality, Transient and steady state solutions of D.C. and A.C. networks.

Network Theorems: Superposition, Thevenin, Norton, Reciprocity. Maximum power transfer. Millman and Tellegen's Theorems and their applications to D.C. and A.C. circuits

Resonance: Resonance in series and parallel- circuits. Q-Factor, bandwidth and selectivity.

Non-sinusoidal Periodic Waveforms: Fourier series- trigonometric and exponential forms. Response of network to non-sinusoidal periodic waveforms.

Two Port Networks: Different two port parameters and their inter-relations and characteristic functions, interconnection of two port networks, Brune's test. Network configurations. Symmetrical and asymmetrical two port communication networks. Iterative, Image and characteristic impedances, Propagation, attenuation, phase and Image transfer constants. Balanced, unbalanced and reciprocal networks; T.L. lattice and bridged T network

Network Functions: Generalized concept of complex frequency, Impedance and admittance functions. Exponential excitation and system functions. Driving point and transfer functions. Pole zero configuration of system functions.

EC 207 A - SOLID STATE ELECTRONICS AND DEVICES (EC)

2L, 1T

3 Hours, 100 Marks

Quantum Mechanics: Review of Schrodinger wave equation. Physical interpretation of wave function, Maxwell Boltzmann, Bose-Einstein and Fermi-Dirac statistics. Interatomic forces and types of bonding. Theory of band structure and electronic states in conductors, semi-conductors and insulators. Fermi level

Semi-conductor Physics: Conduction mechanism in semiconductors. Concept of mobility and diffusion. Concentration of carriers and ionized impurities in non-degenerate semi-conductors. Evaluation of Fermi level. Concept of effective electron mass and hole mass. Carrier transport mechanism in semi-conductors. Drift and diffusion currents in non-degenerate semi-conductors. Thermal equilibrium and non-equilibrium conditions. Carrier recombination, continuity equation and time dependent diffusion equation. Surface states and surface recombination. Carrier life time, Photo conductivity. Introduction to optical absorption, photo luminescence. Cathode luminescence and electro luminescence.

Introduction to Solid State Devices: Space charge region and junction capacitance. Minority carrier injection, carrier storage and transient response. Impact ionization and avalanche break-down.

Analytical theory of junction diodes. BJT, JFET, MOSFET, UJT, diffused transistors, avalanche transistors. Degenerate semi-conductors and theory of tunneling, Theory of tunnel diodes, zener diodes, varactor diodes, photo diodes, LEDs, photo-transistors, photo FETs and LASER. Introduction to Laser. Elementary theory of composite junction. Ohmic junctions and hetero junctions.

EC 204 A - ELECTRONIC CIRCUITS (EC) - I

3L, 1T

3 Hours, 100 Marks

Introduction: Biasing and stabilization techniques of BJT, JFET, MOSFET for use as amplifiers in various configurations. Small signal models for BJT, JFET and MOSFET in discrete and integrated form. Frequency dependence characterization and equivalent circuits. Miller effect.

Untuned small signal BJT amplifiers: Analysis and design of Single stage and multistage. RC coupled and transformer coupled amplifiers. Frequency response, bandwidth, gain and factors affecting them. Various two transistor integrated circuit amplifier stages. Introduction of d.c. amplifiers, differential amplifiers, Cascode and Darlington circuits. Follower circuits and boot-strapping.

Feedback amplifiers: Study of the effects of feedback on amplifier gain, distortion, noise, band-width and impedance levels, Regeneration and its control in multistage audio frequency amplifiers

EC 205 A - DIGITAL ELECTRONICS (EC/ECC/EEE) - I

3L, IT

3 Hours, 100 Marks

Device Characteristics: Steady state and transient switching characteristics of diodes, BJTs, FETs, UJTs. Wave shaping circuits. Integrating and differentiating circuits, effects of time constant, relation of tilt time to time constants. Clipper and clamper circuits using diodes and transistors. Saturated and unsaturated transistor switches. Speed-up capacitors. Inverter circuits. Performance of pulse transformer and lumped distributed parameter electromagnetic delay lines.

Relaxation Oscillators: Theory, operation and performance of astable, monostable and bistable multivibrators. Different triggering circuits. Theory of Schmitt trigger. Comparison of performance of various circuits configurations of multivibrators and their fields of applications. Tunnel diode. UJT relaxation oscillator. Theory of astable and monostable blocking oscillators and their triggering methods.

Sweep Circuits: Free running and triggered modes. Theory and common circuits of voltage and current time base generators.

Sampling Gates: Theory, operation and applications of unidirectional and bi-directional sampling gates using diodes and transistors.

EC 206 A - ELECTRONIC INSTRUMENTS AND MEASUREMENTS (EC)

2L, IT

3 Hours, 100 Marks

Introduction: Difference in measurements at low and high frequencies. Types of indicating instruments, balance detectors and other auxiliary apparatus used. Shielding and grounding considerations. Noise problems. Effects of physical size of components.

Characteristics of Lumped passive components: Measurement of resistance, capacitance, self and mutual inductance and incremental inductance at audio and radio frequencies. Price's guard wire and loss of charge methods for resistance measurement. Important a.c. bridges for capacitance and inductance measurements. Introduction to inductively coupled ratio arm bridge. Special considerations for radio frequency bridges. Twin-T and bridged-T networks. Resonance and heterodyne methods of parameter measurement. Q-measurements. Q-meter.

Instrument Amplifiers: Operation, performance and characteristics of single ended, differential, chopper stabilized amplifiers used in Instrumentation, Different types of choppers and their characteristics .

Electronic Analog Voltmeters: Characteristic and specification of analog electronic voltmeters of different kinds, Circuits of dc. volt meters using tubes, FETs, BJTs, and ICs, Analysis of circuits with various instrument amplifiers, configurations to compare sensitivity, stability, linearity and impedance characteristics. Theory of operation and circuits for average, peak, peak to peak and RMS responding-A.C., Voltmeters, Use of compensated multipliers, CMRR and NMRR, R.F. Voltmeters. Common types of voltmeter probes. Electronic VOM.

Electronic Analog Ammeters: Performance-specifications of instruments for audio and radio frequency current measurements. Rectifier and thermocouple ammeters. Principle of Hall effect ammeters. Use of amplifier type d.c. voltmeter as ammeter.

EC 215 B - LOGIC SYSTEM DESIGN (EC) - I

2T

100 Marks

Number systems: Binary arithmetic, Octal and hexadecimal number systems. Different numerical and alphanumeric codes. Truth functions

Basic Logic Circuits: Positive and negative logic of OR, AND, NOT, NOR, NAND, Exclusive OR and Exclusive NOR gates. RTL, DTL, DCTL, TTL, ECL, MOS and CMOS logic circuits and their realization. Power dissipation, speed, delays, noise margin, fan-in and fan-out capabilities.

Combinational Logic: Boolean algebra. Boolean functions and expressions. Simplification and minimization techniques, K-map and tabular methods. Design of minimal combinational systems and realization. Design of multiple output combinational systems. Design of hazard free combinational system.

Flip-Flops: Unclocked and clocked flip-flops. R-S, D, J-K and T flip-flops. Master-slave and edge-triggered flip-flops. Specifying parameters and timing properties of flip flops .

FOURTH SEMESTER

Ma 251 A - ADVANCED ENGINEERING MATHEMATICS (EC/ECC/EEE) - II

2L

3 Hours, 100 Marks

Section A

Transforms: Laplace Transform, Inverse Laplace Transform, Properties of Laplace Transforms, Application of Laplace Transform to solve differential equation with constant coefficients. Z- Transforms. Infinite Fourier Transforms.

Section B

Numerical Analysis : Interpolation with equal intervals: Newton-Gregory interpolation formulae, Lagrange's interpolation formula for unequal intervals. Central difference interpolation formulae: Gauss' forward and backward formulae, Stirling's and Bessel's interpolation formulae. Numerical integration : Trapezoidal rule, Simpson's 1/3 and 3/8 rule. Numerical solution of algebraic and transcendental equations : Bisection, regula falsi and Newton-Raphson methods. Numerical solution of linear simultaneous equations : Gauss' elimination, Gauss-Jordan, Jacobi and Gauss-Siedal methods. Numerical solution of ordinary differential equations : Euler's, Runge-Kutta Fourth order and Milne's methods.

Section C

Special function : Series solution of Bessel and Legendre's differential equations. Generating function of Bessel and Legendre's Polynomials. Orthogonal Property of

Bessel and Legendre's function. Rodrigue's formula.

Note: Candidates are required to attempt FIVE questions in all, selecting at least one from each Section.

EC 252 A - RF CIRCUITS AND SYSTEMS (EC)

2L, IT

3 Hours, 100 Marks

Coupling Circuits and Impedance Matching Networks: Resistive, capacitive, inductive, mutual inductive and combination type of R.F. couplings. Single and double tuned couplings. Effect of loose, tight and optimum couplings, load conditions and Q. Various impedance matching networks for low and high impedance loads. Balanced-to-unbalanced network.

RF Voltage Amplifiers: Theory, operation, performance and design of R.C. coupled and transformer coupled single and double tuned amplifiers using JFETs, MOSFETs & BJTs. Time and frequency domain analysis. Synchronously tuned multistage amplifiers. Stagger tuned, stagger damped circuits. Neutralization circuits. Cascode amplifiers. Integrated circuit frequency selective R.F. and IF amplifiers

R.F. Power Amplifiers: Special features of R.F. power devices. Theory and design of tuned class C amplifiers using VTs, BJTs, JFETs and MOSFETs. Graphical and approximate analytic methods of solution : Output circuit efficiency considerations. Biasing circuits. Characteristics of driver and output circuit power supplies. Effect of input circuit distortion. Neutralization Parasitic oscillations. Class C amplifiers, adjustments. Theory of frequency multipliers. Tuned class-B linear amplifiers

Sinusoidal Oscillators: Oscillators with two terminal and four terminal active elements. Criteria for oscillations. Analysis, characteristics and applications of different LC, RC, crystal and negative resistance audio and radio frequency oscillators using BJTs, FETs and ICs. Linear and non-linear operation of oscillators. Amplitude and frequency stabilization.

EC 254 A - ELECTRONIC CIRCUITS (EC/EEE) - II

3L, 1T

3 Hours, 100 Marks

Untuned large signal amplifiers: Methods of analysis of large signal FET, BJT and IC amplifiers. Analysis of distortion and cross modulation. Classification of power amplifiers. Analysis and design of single ended, parallel and push-pull Class A, AB and B power amplifiers. Complementary, symmetry and quasi-complementary circuits, Driver and out-put stages, with and without out-put transformers for power amplifiers. Output circuit efficiency calculations for various classes and configuration of amplifiers. Power out-put. Thermal considerations. Derating curves.

Power Supply Circuit: Design factors and applications of various power supply filters and voltage multiplying rectifier circuits.

Regulated Power Supplies: Regulator circuits using voltage regulating tubes, solid state devices and monolithic ICS. Adjustable constant voltage power supplies. Adjustable constant current power supplies. Higher output power supplies with solid state pre-regulations. Protection circuits for power supplies. Rating and specifications.

EC 255 A - DIGITAL ELECTRONICS (EC/ECC/EEE) - II

2L, 1T

3 Hours, 100 Marks

Counters and Registers: Binary and decade ripple counters. One bit counters. Up, down and up-down, synchronous counters. Programmable counters. Divide-by N counters. Storage registers, shift-right, shift-left and bi-directional shift registers. Serial input, serial output, parallel input, parallel output, parallel-in-serial out, serial in parallel out and Universal shift registers and synchronous parallel loading of shift registers. Static and dynamic MOS shift registers. Ring and Johnson counters

Arithmetic Circuits: Digital Comparators, half, and full adders; parallel and serial binary adders, half and full binary subtractors. BCD adders and subtractors. Binary multipliers and divider circuits

Miscellaneous Sub-systems: Encoders, decoders and code converters. Parity checking circuits. Multiplexers and demultiplexers. Digital to analog and analog to digital converters.

Semi-conductor memories: Random and sequential access memories. RAM, ROM, PROM, EPROM, EEPROM, EAPROM, EPLA, GALs. MOS and CMOS memories

EC 257 A - INDUSTRIAL ELECTRONICS (EC/EEE)

3L, 1T

3 Hours, 100 Marks

Power electronics devices: Characteristics and operation of SCR, PUT, SUS, SBS, SCS, TRIAC, DIAC, IGBT, GTO, MCA and light activated thyristors. Ratings and rating extension by series/parallel operation.

Electronic Power Control: Electronic methods of power control. SCR firing methods, Phase control techniques. Line commutation and different types of commutation. One, two and four quadrant converters. Bridge inverters, series and parallel inverters. Cyclo converters

Electrical Drives: Performance characteristics of series, shunt and compound d.c. motors. Motor starters. Characteristics of single and three phase induction motors, Universal motor, Amplidyne and selsyns.

AC and DC Motor Speed Control: Philosophy of speed control, open and closed control and single and three phase AC, DC and universal motors using thyristors. PWM inverter technique and introduction to variable frequency drive

Misc. Industrial Applications: Photo relays and their applications, X-ray tubes. Particle accelerators. Principle of Electron Microscope, Uninterruptible supplies. Switched mode power supplies

Programmable Logic Controllers (PLC): Advantages of PLC, CPU configurations, Digital and analog inputs and outputs, ladder circuits and process flow diagram. Console and operator panel.

EC 265 B - LOGIC SYSTEM DESIGN (EC) - II

2T

100 Marks

Sequential Logic: Storage devices and sequential sub-systems. Introduction to synchronous and asynchronous sequential systems. Mealey and Moore circuits. Cost vs. speed.

Synchronous Sequential Systems: Introductory examples. The finite state model – basic definition. Memory elements and their excitation functions. Synthesis of synchronous sequential circuits. Analysis and design of synchronous sequential circuits. State assignment and reduction technique. Introduction to threshold logic and relay circuits

Asynchronous Sequential Systems: Fundamental mode circuits, Analysis procedure. Circuits with latches. State assignment in Asynchronous sequential circuits. Design of pulse mode asynchronous sequential circuits. Problems in asynchronous circuits – Races and Hazards.

Ma 261 A: Special Mathematics (all branches)

(For Diploma Passed Candidates - common for all branches)

3L,IT

3 Hrs. MM : 100

Differential Calculus: Asymptotes, curvature, envelops evolutes, and curve tracing

Integral Calculus: Rectification. Volumes and surfaces of solids of revolution, differentiations under sign of integration

Differential Equations: Differential equations with constant coefficients and variable coefficients.

Mechanics: Friction , common catenary, kinematics of uniplanar motion, simple harmonic motion

Vector calculus: Gradient , divergence, curl, green's theorem , stoke's theorem, gauss divergence theorem (Verification only)

THIRD B.E. EXAMINATION FIFTH SEMESTER

EC 304 A - ANALOG COMMUNICATION ENGINEERING (EC/ECC)

2L, 1T

3 Hours, 100 Marks

Amplitude Modulation: Analysis of standard AM waves and signal power distribution. Different circuits for amplitude modulation and their comparison. Methods of generating DSBSC SSB and VSB AM and their characteristics. Envelope and coherent demodulation methods for standard AM, DSB-SC, SSB signals. Design considerations of AM modulators and demodulators. Frequency Division Multiplexing.

Angle Modulation: Theory of frequency and phase modulations. Spectrum and BW of FM and PM signals. Direct and indirect methods of generating narrow-band and wide-band FM. Discriminators and PLL demodulators for FM and PM. Pre-emphasis and de-emphasis. Idea of noise suppression properties of FM and PM systems.

Radio Broadcast Transmitters: Characteristics of master oscillators, buffers and frequency multipliers for AM & FM broadcast transmitters. Feeder lines. AM and FM transmitter stability considerations. Broadcast studio equipment and control room apparatus. Stereophonic broadcast considerations.

Radio Receivers: TRF and superheterodyne receivers, Selection of frequency converters, mixers and local oscillators, Characteristics of IF and RF amplifiers. Double spotting. Electrical and Electronic tuning and tracking. Automatic gain and frequency control. Tone control. Band spread. Sensitivity, selectivity, fidelity and noise performance of receivers. Diversity reception. Receiver testing. Special considerations in communication receivers. Double conversion, delayed AGC, noise-limiter and squelch. Operation of AM and FM receivers and their comparison, Introduction to stereo FM receivers.

Pulse Modulation: Uniform sampling theorem, Generation of PAM, PDM and PPM, signals and methods of reconstitution of original signal from its samples. Bandwidth requirements and comparison. Time division multiplexing.

EC 306 A - THEORY AND APPLICATIONS OF INTEGRATED CIRCUITS (EC)

3L, 1T

3 Hours, 100 Marks

Introduction: Distinction between characteristics of linear and digital IC's, Biasing consideration in linear IC's. Parasites and their effects in IC's. General performance, differences between discrete circuits and IC versions. Power requirements and power supplies for linear IC's

IC Fabrication: Processing steps for IC Fabrication, Epitaxial growth, surface passivation, photolithography and etching, elements of diffusion processes. Ion implantation & isolation techniques

Operational Amplifiers: Differential amplifier stage. Characteristics and error signals. Stages of operational amplifiers Multistage operational amplifiers. Characteristic of operational amplifier in different configurations. Frequency response and stability considerations of operational amplifier, Compensation in operational amplifiers. Characteristics of wide band operational amplifiers

Applications of Operational Amplifiers: General applications of Op. Amps. in linear and non-linear circuits. Theory and characteristics of Active filters. Applications of Op. Amps. in wave form generators.

Other Linear ICs: Four quadrant multiplier and its simple applications. CMOS multiplexers. Voltage Regulator ICs with feed back current limiting. Positive- and Negative voltage IC regulators

Linear PLL: Basic theory of first, second and higher order loops. Lock-in and Lock-out processes. Tracking performance. Noise in linear PLL systems. Important applications of linear PLL. Measuring PLL parameters.

EC 309 A - COMPUTER ORGANISATION (EC/ECC/EEE)

2L, IT

3 Hours, 100 Marks

Elements of Computer Organisation

Evolution of Computers: Generations of Computers, Modeling of computers at Gates, Registers and Processors level.

CPU Architecture: Fixed and floating point arithmetic and ALU organization. Instruction format, types, sequencing and interpretation. Instruction fetch and execute cycles. Addressing techniques. Hardwired and Micro-programmed control

Secondary Memories: Magnetic memories core, tape, disk and floppy disc. Introduction to Magnetic Bubbles and CCD memories

I/O Devices: Principle and construction of Keyboard, Mouse, digitizer, joystick, optical scanner, Resistive membrane touch screen, Tele-typewriter, CRT terminals, TFT monitors, Line, Dot Matrix, Daisy wheel, Ink jet and Laser printers.

Communication of I/O with CPU and Memory: Speed mismatch of I/O v/s memory and CPU. Communication methods for I/O to CPU and Memory : Polling, interrupt, DMA and I/O channel.

EC 310 A - RADIATION & WAVE PROPAGATION (EC) – I

3L, IT

3 Hours, 100 Marks

Transmission Lines: Types of transmission lines. General transmission line equations. Line constants and equivalent circuits. Infinite line, Lines with reflections. SWR Lines, dissipationless lines. Cable circuits and composite lines.

Coaxial cables. Transmission lines at audio and radio frequencies. Losses in transmission lines. Transmission equalizers. Characteristics of quarter wave, half wave and other lengths. Smith chart and its applications. Transmission line applications. Stub matching

Radio Wave Propagation: Mechanism of radio wave propagation, Reflection, refraction interference and diffraction of radio waves. Theory of ground wave, space wave and sky wave propagation. Effect of conductivity, dielectric constant, curvature and surface imperfections of earth on wave propagation, Duct propagation and tropospheric scattering. Characteristics of ionosphere and its effects on wave propagation, critical frequency, skipzone and maximum usable frequency. Multiple hop transmission, oblique and vertical incidence transmission. Effect of earth's magnetic field, solar activity and meteorological conditions on wave propagations.

Coordinate System: Cartesian, Cylindrical and Spherical Coordinate Systems. Vector components and unit vectors in the coordinate systems. Gradient, Divergence and Curl in various coordinate systems.

Electric & Magnetic fields: Electric field intensity, Gauss's law, Divergence theorem. Potential difference & potential. Potential Gradient. Conductor Properties & boundary conditions for conductors. Method of Images. Boundary conditions for perfect Dielectric Materials. Biot-Savart Law, Ampere's Circuital law, Curl, Stoke's theorem. Magnetic flux density, Boundary conditions for Magnetic fields.

Vector Magnetic Potential. Faraday's law, Displacement current. Maxwell's equations in point form & Integral form.

EC 311 A - FEEDBACK & CONTROL SYSTEMS (EC)

2L, IT

3 Hours, 100 Mark

Introduction: Elements of control systems, concept of open loop and closed loop systems. Examples and application of open loop and closed loop systems, brief idea of multivariable control systems. Mathematical models and electrical analogy for physical systems. Transfer function analysis. Block diagram, signal flow graphs, reduction techniques and gain formulae.

System Response: Steady state response to DC and periodic functions by pole zero plots. Natural frequencies of networks and transient response. Magnitude and phase from s-plane vectors. Response of linear systems to singularity functions like impulse, step, ramp, parabolic and combination thereof .

System Performance and Stability :Time and frequency response of control systems. Bode plots. Time and frequency response of multiple feedback amplifiers. Proportional integral and Derivative Control, Time domain and frequency domain specification and their correlation.Steady state error coefficient and error analysis.

System Stability: Stability and oscillations of feedback control systems. Concept of absolute, relative and conditional stabilities.Routh-Hurwitz and Nyquist criteria of stability Gain and phase margins. M and N circles and Nichol's chart Root locus technique and its applications. Concept of BIBO and Asymptotic stabilities.

Compensation: Lag, Lead and Lag-Lead compensators Cascade compensation in time and frequency domain. Feed-back compensation techniques.

SIXTH SEMESTER

EC 352 A - ELECTRONIC MEASUREMENT (EC)

2L, 1T

3 Hours, 100 Marks

Power Measurements: Power measurements at audio and radio frequencies. Absorption, calorimetric and transmission power meters. Electronic wattmeters.

Waveform Display Instruments and Recorders: Cathode ray Oscilloscope and its specifications. Cathode ray tube. CRO applications for low and high frequency signals. Characteristics of CRT amplifiers. Free-running and triggered mode, Synchronisation, Dual-beam and dual trace CRO's. Different sweep modes. Mesh storage and phosphor storage CRT's,. CRO Probes,

Strip chart and X-Y servo recorders.

Frequency and Phase Measurement: Frequency standards, Methods of audio and radio frequency measurements, comparisons by CRO and interpolation methods. Absorption, reaction and transmission wave meters. Dip meters. Heterodyne frequency meters, CRO methods of phase measurements, Direct reading, Phase meters for audio and radio frequencies.

Theory of Errors: Accuracy and precision. Methods of reducing systematic errors in measurements. Statistical analysis of random errors. Normal error curve. Estimation of error in computed results.

Amplifier Measurements: Measurements of modulation depth. Different modulation meters. Distortion analyzers and total distortion meters, Intermodulation method of distortion measurements. Square wave testing of tuned and untuned amplifiers. Testing of Video amplifiers

Noise Measurements: Measurements of noise voltage, noise current, noise power and figure. Brute-force method, Sine-wave method and noise generator method. Common types of noise signal generator.

Special Measurements and Tests: Introduction to Sampling, Vector and Differential Voltmeter, Vector impedance meter, Function generators, RF Signal generators and Pulse generators. Introduction to frequency synthesizers.

Wave Measurements: Various types of wave and spectrum analyser for audio, radio and microwave frequencies. Field strength measurements of radio waves.

EC 353 A - DIGITAL COMMUNICATION SYSTEM (EC/ECC)

3L, 1T

3 Hours, 100 Marks

Random signals: Power and energy signals. Introduction to probabilistic and statistical description of discrete and continuous communication processes. Marginal, conditional and joint probability density and distribution functions. Stationarity and ergodicity. Auto correlation and Cross correlation functions. Energy spectral density and Power spectral density. Simple linear system analysis under random excitation in time and frequency domains.

Introduction to techniques ASK, FSK, PSK, BPSK, QPSK and simple QAM.

Digital communication: Comparison of analog and digital communication systems. Essentials of PCM, linear delta and adaptive delta modulation. Study of components of complete digital communication system. Multiplexing, Introduction to bit, word and frame synchronizations. Important digital carrier modulation methods, Introduction to matched filter detection.

Noise: Various noise sources in amplifiers and communication systems. Comparison of various electronic devices for noise performance. Signal to noise ratio and noise figure. Equivalent noise bandwidth. Noise temperature. Effect of cascading, statistical properties of noise. Representation of white and band pass noise in communication systems.

System Performance: Noise-performance of analog CW and pulse modulation systems using coherent and non-coherent detection. Baseband PCM and delta modulation systems performance in terms of probability of error and S/N ratio. Probability of error performance of band pass systems.

EC 358 A - MICROPROCESSORS AND MICROCOMPUTERS (EC/ECC/EEE)

3L, 1T

3 Hours, 100 Marks

Microprocessor Architecture: Architecture of 8-bit 8085, Z80, 6800 Microprocessors; their instruction sets and addressing modes. Assembly language programming of Intel's 8085 Microprocessor. Introduction to assemblers.

Microprocessor interfacing: Interfacing of address, data and control buses, Memory and I/O devices, Interrupt and DMA for 8085 microprocessor.

Introduction of Microcontrollers: Architecture and instruction set of MCS-51 series of microcontrollers. Application of Microcontrollers.

16 and 32-bit Microprocessors: CPU architecture, addressing modes and features of 16 and 32 bit microprocessors – 8086. Salient features of 80286, 80386, 80486 and Pentium series microprocessors.

Bus standards: Introduction to Multibus, VME, RS-232-C, IEEE 488, PCI, USB, RS 422 and 485.

EC 355 A - RADIATION AND WAVE PROPAGATION (EC) – II

2L, 1T

3 Hours, 100 Marks

Radiation: Retarded potentials and concept of electromagnetic radiation. Alternating current element and power radiated, Radiation resistance. Radiation from dipole and monopole antennas.

Antennas: Quarter wave and half wave antennas. Application of network theorems to antennas. Resonant and non-resonant antennas. Radiation resistance, J impedance, mutual impedance and directional characteristics of antennas. Antennas patterns. Effective length and effective area of antennas, Antenna gain efficiency, beamwidth and polarization. Antenna temperature. Loaded antennas. Colinear, broad-side, endfire and combination arrays and computation of their radiation patterns. Multiplication of radiation patterns. Binomial array, Yagi Rhombic, log periodic antennas and Baluns. Receiving antennas. Antenna systems for diversity reception. Elements of design considerations for medium wave and short wave antennas.

Antenna Measurements: Antenna length, resonance, impedance, gain directivity polarization, phase and radiated power measurements.

EC 361 A – AUDIO & VIDEO ELECTRONICS (EC)

2L, IT

3 Hours, 100 Marks

Audio Systems: Monaural, binaural, monophonic, stereophonic and quadraphonic, high-fidelity and surround-sound audio system descriptions. Digital and optical recording and reproduction.

Television Transmission: Scanning and synchronizing. Analysis of composite video signals. Common standards of Television. Colour TV cameras, Picture carrier, sound carrier and colour carrier signals. Encoding the picture information, Chrominance modulation. Colour resolution. Compatibility of Monochrome and colour systems. Block diagrams of colour TV transmitters and functions of each part.

Television Reception: Decoding the picture information. Functional block diagrams of Monochrome and colour TV receivers. Raster, sync, and AGC circuits. R.F. alignment.

Video Systems: Principle of working of video tape recorders and disc players. Common standards. Introduction to MPEG digital recording.

FINAL B.E. EXAMINATION

SEVENTH SEMESTER

EC 402 A – TELEMATICS (EC)

3L, IT

3 Hours, 100 Marks

Digital Telephony: Principle of working of SPC digital telephone exchanges. Digital switching, space, time. TS, ST, STS, TST switch blocks. Termination of subscriber lines Signalling systems with digital exchanges. Principle of common channel signalling. Synchronization aspect for digital telephony. Store Program Control for call processing.

Integrated Digital Networks: Data Communication terminology. Introduction to Circuits, message and packet switching concept. Basic aspects of multiplexing, signaling and synchronization in integrated digital networks. Overview of ISDN and BISDN. Concepts of basic rate and primary rate ISDN. Access and facilities provisions. Elements of fast packet switching, frame relay, ATM, SONET and SDH. Introduction to photonic switching.

EC 403 A - ADVANCE COMMUNICATION SYSTEMS (EC/ECC/EEE)

3L, IT

3 Hours, 100 Marks

Line-of-Sight and Troposcatter Communication: Principle of working and essential features of microwave LOS and troposcatter communication. Propagation study and performance requirements. Diversity techniques.

Satellite Communications: Basic considerations. Up-link and down link parameters. Orbit and frequency selection. Transmission losses, noise and interference. Elements of multiple access techniques. Frequency reuse techniques. Functional description of earth stations

Optical Communications: Ray propagation in optical fibers. Types of fibers. Losses and dispersion in fibers. Transmitter and receiver subsystem for optical communications. Laser and LED sources. Optical Amplifier. Cable joints, couplers and connectors. Splicing techniques. Modulation techniques. PIN and avalanche photo diode detectors. Characteristics of analog and digital transmission in optical communication systems. Noise considerations.

EC 407 A – RADAR AND NAVIGATIONAL AIDS (EC)

3L, IT

3 Hours, 100 Marks

Radar Operation and Performance: Components of radar system Radar performance factors and specifications. Pulse considerations and duty cycle. Equations for monostatic and bistatic radar, Radar cross section and range. Pulse integration. System losses. Radar antenna patterns, phased arrays and interferometer. Duplexer, Radar tracking and scanning. Monopulse, sequential lobing and conical scanning methods. CW, FMCW, MTI, SAR and Pulse doppler radars. Weather radars

Electronic Navigation: Principle of operation of radar direction finder and range system, Loran system, DME and TACAN system. Microwave altimeters, Elements of Microwave remote sensing. Satellite based Global Positioning System.

EC 404 A - ADVANCED COMPUTER TECHNOLOGY (EC/EEE)

3L, IT

3 Hours, 100 Marks

Computer Architecture: Microprogrammed control organization, CPU-memory speed mismatch and solutions : Word-length, caches and buffers. Speeding up instruction cycle, instruction fetch and decode overlaps.

Parallelism in execution: Instruction pipe-line and vector processing concepts. Introduction to Array processors.

Memory Organisation: Memory hierarchy. Associative memory. cache memory, Paging and segmentation.

Advanced Systems: Organization of parallel processing, multiprocessing, multiprogramming, distributed processing and time sharing systems.

Software: Introductory theory of Compilers, Interpreters and operating systems.

EC 408 A - ENGINEERING MANAGEMENT AND ECONOMICS (EC/ECC/EEE)

2L

3 Hours, 100 Marks

Principles of Management: Management function. Theories of management and their application Span of control, Responsibility, authority, leadership, motivation, communication, Management of Change, Importance of organization structure

Financial Management: Functions and importance of financial management, Book-keeping, Interest and depreciation. Salvage value Various types of costs, profit/volume ratio. Break even analysis and marginal costing

Marketing and Strategic Management: Concept of marketing and its various components Product Life Cycle Strategic Mission Vision Goals Industry Life Cycle SWOT analysis

Stores and Purchase Management : Function of store and Purchase management.

Inventory Management:. Inventory control and management Economic order quantity. A-B-C analysis..

Forms of Business Ownership: Proprietorship, partnership, joint stock companies, joint sectors

Production Planning and Control: Job, Batch and mass production, production efficiency, productivity.

Site selection. Production planning. Routing, scheduling and follow up. Elements of time and motion study. Quality control and quality assurance

Nature and Scope of Economics: Basic concept of managerial economics. Supply and demand, free competition, monopoly and oligopoly

EIGHTH SEMESTER

EC 451 A - DIGITAL SIGNAL PROCESSING (EC/EEE)

2L, IT

3 Hours, 100 Marks

Digital Signal Processing: Advantages of digital filters and processing. Fundamentals of discrete time systems. Fourier transform of sequences. Discrete-time filter structures. Z-transform system representation, solution of linear constant co-efficient difference equations. Digital filters design by transformation from analog filters. Simple realizations of IIR and FIR filters. DFT and FFT.

EC 452 A - ELECTRONIC INSTRUMENTATION (EC)

2L, IT

3 Hours, 100 Marks

Transducers: Construction, characteristics and circuits for common types of resistive, capacitive, inductive, magneto-structive; piezo-electric. Photo-electric and thermo-electric transducers for measurement of process physical variables. Various sensing elements and transducers for measurement of Force, Pressure, Humidity, Moisture, strain, Velocity, Acceleration and pH. Inductive and Capacitive proximity switches. Physical and electrical loading of and by the transducer Systems.

Signal Conditioning: Analog and digital signal conditioning for instrumentation. Applications of OPAMP, A/D and D/A converters. Use of microprocessors. Techniques of S/N ratio improvement, Methods of shielding and grounding. Protection from RFI & EMI.

Electronic Displays: Principle of LED numeric, matrix and alpha-numeric displays, gas discharged plasma panels, flat panel CRT, LCD, electro-luminescent and electrophoretic displays.

Digital Instrumentation: Principle of operation of probes, logic monitors, digital pulser and logic analyser. Components of a digital universal counter. Digital period; frequency and time interval measurement. Errors in digital counter. High frequency measurement with digital counters. Different types of integrating and non-integrating digital voltmeters and multimeters, Errors and field applications of various digital voltmeters and counters. Introduction to Digital oscilloscope. Elements of data acquisition systems.

EC 457 A - MOBILE COMMUNICATION (EC)

2L, IT

3 Hours, 100 Marks

Basic technical concepts: Concept of mobile telecommunications. Mobile radio network issues, cell size, coding, modulation and diversity. Base station subsystems. Access methods. Location strategies for personal communications services. Cell design principles.

Radio Paging System: Types of radio paging system. On-site and wide area paging, digital codes like POCSAG, elements of radio paging system engineering.

Microcellular radio communications: Fixed and dynamic channel assignment. Cellular systems and power control. Basic features and system architecture in cordless telephones. Marine and aircraft communication. Overview of mobile terrestrial communication by satellite.

Cellular Radio System: Allocation of spectrum and channels, Concept of Hexagon cells, mobile identification system and registration of mobile, call procedure and measurement of signal strength, GSM and CDMA radio system architecture, roaming, digital speech and channel coding. Efficient use of radio spectrum, multi operator working, cells and frequency reuse.

EC 458 A - COMPUTER COMMUNICATION AND DATA NETWORKS (EC/ECC/EEE)

3L, IT

3 Hours, 100 Marks

Computer Communication: Layered architecture of computer communication networks. DNA, SNA and ISO-OSI models. Properties of LANs, MANs and WANs. Physical level, data link and transport protocols. Multiple access protocol organization. Routing techniques, flow and congestion control in packet switched networks. Window scheme. Network interconnection-bridges and routers. Dead lock avoidance. Elements of queuing analysis. Introduction to network security.

Data Networks: Structure and functions of network protocols. Data link control procedures. Operation of HDLC, SDLC, BISYNC, X.25 and X.21 Protocols Elements of Polling, ALOHA, Reservation ALOHA, CSMA and token ring. Characteristic features of LANs

Basics of Internet: Evolution; dialup, XDSL, ADSL, cable modem and other access methods. IP address and domain name system, TCP/IP, Internet applications and www.

EC 459 A - MICROWAVE ENGINEERING (EC/EEE)

3L, IT

3 Hours, 100 Marks

Wave Guides: Theory of wave propagation in rectangular wave guides, cut off frequency. Dominant and higher modes. Generation of different modes and suppression of unwanted modes. Field distribution. SWR and impedance relations in wave guides. Coupling between coaxial lines and wave guides. Wave guide stub-matching

Resonators: Theory and application of cavity resonators. Coupling to cavity, Q of cavity resonators

Microwave Components: Details about attenuators, phase shifters, directional couplers, tees, isolators, circulators, tuning, screws, coupling probe, loops, mixers and detectors. Use of scattering parameters

Microwave Generators and Amplifiers: Theory of velocity modulation. Theory of operation and characteristics of two cavity and multicavity klystron, amplifier and oscillators. Reflex klystron. O and M type travelling wave tube and backward wave oscillators - principle of operation. Construction, type and application of Magnetrons.

Microwave Solid State Devices: Special considerations for UHF and microwave transistors and oscillators. Parametric amplifiers. Manley-Rowe relation, linearized equations. Parametric up converters. Negative resistance amplifiers. Principle of working and application of impact diode, hot carrier diode, PIN diode, Gunn diode and LSA diode. Quantum mechanical explanation, description and application of MASER amplifiers.

ELECTIVES

EC 481 A - MEDICAL ELECTRONICS (EC)

2L, IT

3 Hours, 100 Marks

Introduction to Physiology: Physiological system of the human body. Nerve physiology. Function of nerves and myoneural junction. Cardiac muscle and its contractions. Blood flow system. Arterial pressure. Mechanism of respiration, Function of spinal cord. Generation, propagation and distribution of action potentials.

Recording of Bio-electric Events: Kinds of electrodes, amplifiers and display units for recording bioelectric potentials. Principles of ECG, EEG, and EMG. Electrophysiological signals from a micro electrode and salt bridge, Use of field effect-devices as electrometers. Principle of driven shield. Use of photon-coupled amplifiers. Artifacts

Bio-Medical Measurement: Electronic methods of measuring blood pressure, blood flow, blood pH, skin and systemic body temperature and pulse rate.

Electronic Medical Instrument: Electronic pace makers. Implantable power sources. Defibrillators. Micro power transmitter for telemetering bio-signal. Surgical and therapeutic diathermy units. Physiological stimulators. Basic diagnostic X-ray units. Introduction to patient monitoring and intensive care units.

EC 482 A - DESIGN PRINCIPLES OF COMMUNICATION SYSTEMS (EC)

2L, IT

3 Hours, 100 Marks

Design methodology: Design methodology for AM and FM radio broadcast, transmitters and receivers.

Design aspects of PCM: PCM transmitters and receivers with base band transmission.

Design considerations for microwave and optical communication: analog and digital microwave communication. LOS links, troposcatter links and optical communication links

EC 484 A -SOLID STATE AND IC TECHNOLOGY (EC)

2L, IT

3 Hours, 100 Marks

Solid State Device Fabrication: Crystal growth and doping. Impurity control and junction formation methods. Zone processes, Pick's laws, Evaluation of diffused layers. Epitaxial- systems and processes

Fabrication considerations: Diffusion during subsequent processing. Oxidation techniques and oxide masking. Etching techniques and processes, Photo-engraving. Contacts and interconnections. Alloying Epitaxial planar technology. Packaging, Scaling, Introduction to the technology of semiconductor devices using III-V & II-VI groups compounds

IC Fabrication: -Epitaxial growth. Surface passivation techniques. Photolithography. Etching. Junction formation. Diffusion processes. Constant source and limited source diffusion. Ion implantation. Isolation techniques. Monolithic diodes, transistors, capacitors and resistors. Differences in bipolar and MOS-ICs. Introduction to thin and thick film techniques

EC 485 A - COMPUTER AND MICROPROCESSOR CONTROL OF INDUSTRIAL PROCESSES (EC)

2L, IT

3 Hours, 100 Marks

Process Control Computer Systems: Minis, Micros classification by hardware features and software facilities. Performance evaluation techniques. System selection criteria

Characteristics of Digital Processors: Organisation and characteristics for process control. Input/output arrangements. Addressing techniques. Memory systems

Device Technology and Process Environment: Assessment of the use of digital switching devices in hazardous environments—Noise, temperature, humidity, vibration-static and dynamic.

Transducers: Generation of digital signal using process transducers, Multiplexing

Process Control System Software: Introduction to process control languages, application packages and operating system for real time process control. Development systems for Micros Introduction to logic analyser, cross assembler and cross compilers

EC 487 A - ENTREPRENEURSHIP DEVELOPMENT (EC)

2L, IT

3 Hours, 100 Marks

Entrepreneurship and Economic Development: Definition of entrepreneurship. Entrepreneurship and economic growth. History of Socio-Economic development in India. Achieving societies and their development statistics. Characteristics of under-developed, developing and developed societies, Wage employment and self-employment, Linkage between large, medium and small scale industries.

Entrepreneurship Characteristics and Skills: Identification of personal characteristics of an entrepreneur. Goal setting, risk taking, decision-making, time management and problem solving. Staff assessment

Creativity: Hindrance to creative thinking, developing creativity, personality traits of creative persons

Leadership: Qualities and influencing abilities. Negotiation skills and their development

Communication: Types of communication, barriers to effective communication, methods for improving communication

Entrepreneurship Opportunities and Selection: Types of entrepreneurship opportunities—manufacturing, service, trading and professional, Need analysis. Identification of local resources

Evaluation of Potential Opportunities: Setting decision criterion and selection of product/service/profession. Evaluation Technical, Financial and Economic feasibility

Starting a New Enterprise: Nature of information necessary for getting started. Information from governmental agencies like, SISI, RICO, RFC, DIC etc. Procedural requirements for establishing new enterprise, Selecting type of business organisation. Registration and Licensing, Incentive and Facilities available for small scale enterprises.

Enterprise Management: Product Design and Development Concepts of product design and development. Design by innovation and evolution. Enhancing product value

Facilities Planning and Production Management: Selection of relevant technology. Make or buy decisions. Plant layout and process planning, setting production levels and delivery schedules, Economic utilisation of

labour, raw material and equipment. Inventory Management. Control of production plants. Quality assurance and quality circle

Product Costing: Fixed costs and variable costs. Sales price and revenue. Break even analysis. Product Pricing. Profit planning

Financial Management: Financial Management and accounting in small enterprise, Difference between financial, accounting, cost accounting and management accounting, Book-keeping, Balance sheet, "Profit and Loss Account and their analysis. Cash flow and financial ratio". Assessing financial needs of enterprise. Fixed and Working Capital, Sources of funding

Financial Regulation and Taxes: Sales Tax Act, Income Tax Act, Excise, Local Taxes and Insurance

Practical Work: Students be given practical work in identification of opportunities, need analysis, market survey, evaluation of alternatives and preparation of feasibility report

EC 488 A – RELIABILITY AND MAINTAINABILITY OF ELECTRONIC SYSTEMS (EC)

2L, IT

3 Hours, 100 Marks

Introduction: Life cycle, Reliability and Maintainability management, design reviews.

Reliability: Mathematical modeling, environmental factors, fault tree analysis, failure reporting and corrective actions, reliability testing.

Maintainability: Mathematical modeling, system design, unit design, assembly design, maintainability testing, reliability and maintainability documents.

EC 490 A- ROBOTICS ENGINEERING (EC)

2L, 1T

3 Hours, 100 Marks

Introduction: Introduction to Sensors, drivers, isolators, actuators, microprocessors & microcontrollers, DC and DAC. Basic concepts of pneumatic, hydraulic, and mechanical movements. Basic Robot elements, Types of Robots, Industrial applications, Future trends.

Sensors: Ultrasonic, temperature, stress & strain, optical, displacement, gravity & tilt sensors. Introduction to robotic vision and image processing.

Robot output: Robot output control, stepper motor, servo motors, DC motors and drives. Grippers, transporters, rotators, and cable drive systems.

Robotics: Need and application of sensors in robots, Sensor interfacing, data acquisition and conversion systems, encoders and decoder applications in robots, microcontroller/microprocessor interfacing for sensors and actuators, Robot Axes: introduction to different robotic axes systems and their drive configuration. Robotic control system, feedback mechanism. Introduction to Robot dynamics and application of robots in industry. Concept of autonomous automata robots.